Programming Techniques –Using C Language

Objective: The purpose of this lab is to familiarize you with C variables: variable declaration and initialization. Also to read data from the standard input device and store it in a variable and print it to the standard output device.

Declaring and Printing variables

1. Write the following C program, execute it and check the output.

```
#include <stdio.h>
int main(){
    int a;
    int b;
    int sum;
    a=20;
    b=10;
    sum=a+b;
    printf("Summation of a and b=%d\n",sum);
    return 0;
}
```

Figure 01

2. Execute C programs given in Figure 02 and 03. Check the output of the programs.

```
#include <stdio.h>
main(){
    int n;
    float m;
    char c='A';
    n=5;
    m=3.5;
    printf("Number=%d\n",n);
    printf("Floating point number=%f\n",m);
    printf("Character=%c\n",c);
    return 0;
}
```

Figure 02

```
#include <stdio.h>
main(){
        int luckyNo = 5;
        float radius = 2.0;
        char myName[15] = "John";
        char initial = 'J';
        printf("Hello World\n");
        printf("My lucky number is %d\n", luckyNo);
        printf("My name is %s\n",myName);
        printf("My first initial is %c\n",initial);
        printf("The radius of a circle is %f\n", radius);
        printf("Hello %s or should I say %c\n",myName,initial);
        return 0;
}
```

Figure 03

3. Create a program called "c_variables.c" to declare group of character variables called char1, char2, char3. Assign characters 'A',' B', 'C' to these character variables. Eg. char1='A'. Write a single printf() statement to get the output as Figure 04.

```
ABC  reversed is CBA
```

Figure 04

<u>Integer and floats can be displayed in different formats in C programming.</u>

4. Execute C program given in Figure 05. Check the output of the program.

```c
#include <stdio.h>
int main(){
        int integer = 9876;
        float decimal = 987.6543;
        printf("4 digit integer right justified to 8 column: %8d\n", integer);
        printf("4 digit integer right justified to 5 column: %5d\n", integer);
        printf("Floating point number rounded to 2 digits: %.2f\n",decimal);
        printf("Floating point number rounded to 0 digits: %.f\n",987.6543);
        printf("Floating point number in exponential form: %e\n",987.6543);
        return 0;
}
```

Figure 05

5. Type given program in Figure 06, correct errors and execute it.

```c
int main (){
    printf ("Arithmetic Operators\n\n");
    i = 6;
    printf ("i = 6, -i is : %d\n", -i);
    printf ("int 1 + 2 = %d\n", 1 + 2);
    printf ("int 5 - 1 = %d\n", 5 - 1)
    printf ("int 5 * 2 = d\n", 5 * 2);
    printf ("\n9 div 4 = 2 remainder 1:\n");
    printf ("int 9 / 4 = %d\n", 9 / 4)
    printf ("int 9 % 4 = %d\n", 9 % 4);
    printf ("double 9 / 4 = %f\n", 9.0 / 4.0);
    return 0;
 }
```

Figure 06

Using simple operators in C

6. Write a program that converts Centigrade to Fahrenheit.

$$F = \frac{9}{5}C + 32$$

7. Write a program to calculate the area of a circle.

$$Area = 2\pi r$$

8. Write a program to calculate the volume of a sphere

$$Vol = \frac{4}{3}\pi r^3$$

Reading data from the standard input device

General format-Reading data from the standard input:

scanf("Control string", &variable);

9. Compile and execute the code given in figure 07.

```
#include<stdio.h>
int main(){
    int number;
    printf("Please enter a number :");
    scanf("%d",&number);
    printf("You entered %d\n",number);
    return 0;
}
```

Figure 07

10. Modify this program to read a floating point number and a character.

11. Write a C program to perform addition, subtraction, multiplication and division of two numbers. Two numbers should read from the keyboard.
12. Write a C program to enter radius of a circle and find its diameter, circumference and area.
13. Write a C program to convert specified days into years, weeks and days.
    Note: Ignore leap year. And read the no of days from the keyboard.
14. Write a program that reads,
    a. An integer as the number of minutes, and outputs the total hours and minutes. (Eg. 90 minutes = 1 hour 30 minutes).
    b. Hours and minutes as input, and then outputs the total number of minutes. (Eg.1 hour 30 minutes = 90 minutes).

15. Write a c program to fulfill the following requirements.
    a. Declare integer variables “x” and ”y”.
    b. Then read data to variable x and y from the keyboard
    c. Calculate values of the following expressions and print them on the screen with appropriate texts.
        i. **x * y**
        ii. **x /y**
        iii. **x % y**
        iv. **x + 3 * x/(y–4)**
        v. **(x+3 * x)/(y–4)**